THE RESTRICTION OF AN M-REGULAR COHERENT SHEAF TO A HYPERPLANE IS M-REGULAR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Cohomology of a closed immersion pushforward agrees with cohomology on the hyperplane. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-relative-affine-cohomology (tag 089W). Lets the cohomology of G be computed after pushing it into projective space.
- The exact sequence of the induction situation and its cohomology sequence. Stacks Project, Varieties, Lemma lemma-exact-sequence-induction (tag 0ABH) and Remark remark-exact-sequence-induction-cohomology (tag 0ABI). Supplies the short exact sequence whose long exact cohomology sequence is the displayed one.

Source reference. Original lemma 08A0: sources/stacks-chow/tags.txt, source lines 10515–10515. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 089W: sources/stacks-chow/tags.txt, source lines 10511–10511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original remark 0EGK: sources/stacks-chow/tags.txt, source lines 18100–18100. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In the situation of Lemma 08A0, if $\mathcal{F}$ is m-regular, then $\mathcal{G}$ is m-regular on $H \cong \mathbf{P}^{n-1}_k$.*

Proof. Recall that $H^i(\mathbf{P}^n_k, i_*\mathcal{G}) = H^i(H, \mathcal{G})$ by Cohomology of Schemes, Lemma 089W. Hence we see that for $i \geq 1$ we get

$$H^i(\mathbf{P}^n_k, \mathcal{F}(m-i)) \to H^i(H, \mathcal{G}(m-i)) \to H^{i+1}(\mathbf{P}^n_k, \mathcal{F}(m-1-i))$$

by Remark 0EGK. The lemma follows. □

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